MILITARY SPECIFICATION

PUMPS, CENTRIFUGAL, SUBMARINE TRIM AND DRAIN

(SERIES TYPE ONLY)

1. SCOPE

1.1 This specification covers the requirements applicable to the design and construction of electric motor driven centrifugal pumps (series type only) for trim and drain service aboard Naval submarines.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball, Annular, (General Purpose).
FF-B-185 - Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
QQ-A-327 - Aluminum Alloy 61S; Plate and Sheet.
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (K-Monel).
QQ-N-288 - Nickel Copper Alloy and Nickel-Copper-Silicon Alloy Castings.
GGG-P-781 - Pullers, Bushing, Bearing, Gear, Wheel Hub and Cylinder Sleeve, Installing and Removing.

MILITARY

JAN-P-735 - Primer, Paint, Zinc-Chromate, Alkyd Type.
MIL-S-901 - Shockproof Equipment, Class HI (High-Impact) Tests for.
MIL-D-963 - Drawings, Production, Electrical and Mechanical Equipment, Naval Shipboard.
MIL-C-2174 - Controllers, Motor Starters, and Master Switches - Direct-Current, Naval Shipboard Use.
MIL-P-15024- Plates, Identification—Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-M-15071 - Manual, Technical, for Mechanical and Electrical Equipment (Less Electronics).
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-R-15109 - Resistors and Rheostats, Shockproof (Naval Shipboard Use).
MIL-R-15137 - Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-C-15328 - Coating, Pretreatment (Formula No. 117 for Metals).
MIL-B-16033 - Bronze, Aluminum: Castings.
MIL-B-16261 - Bronze, Bearing: Castings.
MIL-B-16444 - Bronze, Hydraulic (Ounce Metal): Castings.
MIL-M-16576 - Metal, Gun: Castings.
MIL-P-16789 - Preservation, Packaging, Packing and Marking of Pumps, General, and Associated Repair Parts.
MIL-M-17413 - Motors, Direct-Current, Integral H. P, Naval Shipboard.
MIL-P-17545 - Primer, Paint, Shipboard, Alkyd - Red Lead Type, Formula No. 116.

FSC 4320

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MILITARY (cont'd.)
MIL-B-17931 - Bearings, Ball, Annular, Specially Selected for Quiet Operation (Noise Tested).
MIL-P-18608 - Pumps, Centrifugal, Submarine Trim and Drain.
MIL-C-20159 - Copper-Nickel-Alloy (70-30): Castings.

NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

MILITARY
MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy.

DRAWINGS

BUREAU OF SHIPS
B-153 - Packings and Gaskets, Standard Application of.
810-1, 385, 860 - Flanges, Dimensions, Submarine Pumps.

(Copies of specifications standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services,

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Qualification - Pumps furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein (see 6.2).

3.2 General design -

3.2.1 The principle of reliability is paramount and no compromise of this principle shall be made with any other basic requirements of design. It is the intention of this specification to obtain purpose of such design that they will operate over a long period of years with a minimum of servicing. Where wear or erosion is unavoidable the parts subjected to such wear or erosion shall be of the best materials available for the. purpose in order to reduce these detrimental effects to a minimum. The design and construction of all pumps shall be the most compact consistent with the following requirements.

(a) Reliability.
(b) Accessibility for repair.
(c) Resistance to wear or corrosion
(d) Economy.
(e) Satisfactory operation when inclined as follows:
(1) During submerged operation permanently inclined 30 degrees from the normal position and when rolling 30 degrees from the vertical to either side. The time for a complete roll cycle shall be 12 seconds

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During submerged operation with 15 degrees permanent list to either side of the vertical or when pitching 10 degrees up or down, period for complete pitch cycle 4 seconds.

(3) For surface operations with ship rolling 60 degrees from vertical to either side.

(4) For snorkel operation with ship rolling 30 degrees from vertical to either side.

(5) During surface or snorkel operation with 15 degrees permanent list to either side of the vertical plane and permanently inclined up or down 7 degrees from the horizontal plane. The time for a complete roll cycle shall be considered as 8 seconds on the surface and 12 seconds when snorkeling.

(6) During continuous snorkel or surface operation when ship is pitching 10 degrees up or down from the horizontal plane at a period of 6 seconds for surface and 4 seconds for snorkel operation.

3.2.2 The pump units shall be designed to obtain the quietest possible operation consistent with the requirements specified in 3.2.1. Noise and vibration standards as specified in the Addendum to Purchase Specifications for Machinery - Noise and Vibration shall be complied with. (Copies may be obtained through the Bureau of Ships, Code 549.)

3.2.3 Pumps shall have constantly rising head-capacity characteristic curves. Unless otherwise specified in the. contract or order, pumps shall be designed to operate satisfactorily with a maxim urn suction lift of 25 feet.

3.2.4 Pumps shall be capable of operating at any speed between those required to obtain the mini-m urn and maximum rating points specified (see 6.1).

3.2.5 The pumps shall be multi-stage with single suction impellers designed to give a minimum hydraulic unbalance.

3.2.6 Pumps shall be designed for series operation through the entire range of pressure,

3.2.7 Pumps shall be vertically mounted with the motor at the top, A mounting ring or table shall be provided which will support the unit as near to the center of gravity as practicable.

3.2.8 Shockproofness. -

3.2.8.1 The design of all complete pumping units shall be such that they are capable of passing the ciass HI (hign-impact) shock test specified in Specification MIL-S-901.

3.2.8.2 Pumps shall be designed for vertical mounting by means of a bracket to which the motor and pump casing are attached by rigidly bolted flanges. The entire assembly shall be attached to the bulkhead or other vertical support by means of resilient mountings. Brackets and resilient mountings shall be furnished by the pump manufacturer. Bracket dimensions and data on resilient mountings shall be furnished to the manufacturer by the shipbuilder.

3.2.8.3 The determination of stresses resulting from the shock tests is accomplished by the appli-cation of the usual methods of analytical mechanics. The allowable stress in components due to shock shall be the yield strength (0. 2 percent offset) of the material. Normal stresses in parts, such as the tensile stress in a boil due to initial pull-up, shall not be added to the shock stresses. It is of para-mount importance that the maximum accelerations possible under the shock tests be utilized in the design of holding down bolts, supporting feet or lugs, and main structural members of the equipment, and that no failure or permanent deformation occur as a result of shock.

3.28.4 Bolts designed to be stressed in shear shall be installed in holes with a minimum of clearance, as experience indicates that large clearances allow impacting and subsequent failure.

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3.3 Detail requirements. -

3.3.1 Casings. -

3.3.1.1 The casings shall be split on the shaft centerline so as to permit ready removal of the rotating element and replacement of bearings and glands.

3.3.1.2 Sufficient clearance shall be provided around bolt heads and nuts to permit the use of ordinary tools.

3.3.1.3 Fitted bolts or heavy dowel pins shall be provided to insure maintenance of alignment of casing halves in reassembly.

3.3.1.4 Forcing bolts shall be provided for breaking joints.

3.3.1.5 Pumps shall be fitted with drain connections.

3.3.1.6 Pumps shall be fitted with connections for suction and discharge pressure gauges, conforming to Drawing 810-1,385,860.

3.3.1.7 The pumps will be installed in conjunction with wet vacuum type priming pumps and provision shall be made for receiving priming lines at high joints in the casings.

3.3.2 Suction and discharge connection. -

3.3.2.1 Suction and discharge connections shall be flanged in accordance with Drawing 810-1,385,860.

3.3.3 Impellers and shafts. -

3.3.3.1 Outside surfaces of impellers shall be smooth finished, and inside surfaces shall be file finished insofar as practical.

3.3.3.2 Impellers shall be keyed on the shaft and securely held against lateral movement by locked nuts or other means approved by the bureau or agency concerned.

3.3.3.3 Shaft threading shall be counter to the direction of rotation.

3.3.3.4 Shafts shall be fitted with shaft sleeves in way of stuffing boxes and diaphragms.

3.3.3.5 Shaft flingers shall be provided adjacent to stuffing boxes.

3.3.4 Stuffing boxes and packing. -

3.3.4.1 Stuffing boxes shall be of adequate depth and design to reduce leakage to a minimum under all operating conditions.

3.3.4.2 Packing shall be in accordance with Drawing B-153. All packing shall be of a brand approved by the bureau or agency concerned.

3.3.4.3 All stuffing boxes shall be fitted with throat bushings.

3.3.4.4 The stuffing boxes shall be fitted with water seal connections and a bronze lantern ring located between inner and outer rings of packing. Pumps shall be furnished complete with water seal piping, fittings and valves as required. Special care shall be taken in the design to insure that compression of the inner packing will not allow the lantern rings to move sufficiently to cut off or restrict the sealing water supply.

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3.3.4.5 The stuffing box shall be sealed from the pump discharge or an intermediate stage in a manner satisfactory to the bureau or agency concerned.

3.3.4.6 The design shall be such as to insure that leakage from the glands cannot reach the bearings or be thrown over bearing housings or driving units. This shall preferably be accomplished by using glands cored out internally and the shafts provided with flingers. Ample drip pockets, spray shields and drains shall be provided.

3.3.4.7 Gland and lantern rings shall be split in halves with the halves secured together in a manner to form solid rings for setting up.

3.3.4.8 All stuffing box glands shall be set up by nuts threaded on gland swing bolts, or on studs secured in the casings.

3.3.4.9 Sufficient space shall be provided between bearings and stuffing boxes to permit easy rem oval of packing, and inspection and overhaul of bearings.

3.3.5 Couplings. -

3.3.5.1 A flexible coupling shall be installed between the pump and driving unit.

3.3.5.2 Couplings shall be either bronze, bushed with neoprene or equal, or of an all metal type, providing the design of the coupling is such as to avoid the generation of noise.

3.3.5.3 All coupling hubs shall be keyed to the shafts and secured by lock nuts. For shafts one inch in diameter and larger, the hubs shall be fitted on a taper with keys parallel to the tapers.

3.3.6 Bearings. -

3.3.6.1 A suitable thrust bearing shall be installed at the top of each pump for counteracting any unbalanced hydraulic or mechanical thrust in either direction. In this connection, consideration shall be given to the fact that rolling, pitching or list of a ship at sea may introduce thrust loads even though the unit is in hydraulic balance.

3.3.6.2 Thrust bearings shall be of the combined radial and thrust ball type, They shall preferably by type 134, class I, single row, angular contact, duplex mounted, ball bearings in accordance with Specification FF-B-171. Bearings shall be certified “noiseless” in accordance with Specification MIL-B-17931 by a Government laboratory.

3.3.6.3 The lower radial bearing shall be of the sleeve type.

3.3.6.4 The use of ball bearings shall be subject to the speed limitations of Specifications FF-B-171 and FF-B-185, except where prior approval of the bureau or agency concerned has been granted.

3.3.6.5 Bearings shall be installed in housings separate from and independent of the stuffing boxes with adequate protection from gland leakage. Suitable means shall be provided to prevent escape of oil around the shaft. All baffles, wipers and related parts shall be readily renewable.

3.3.7 Lubrication. -

3.3.7.1 The lubrication of ball bearings shall be in accordance with Specification FF-B-171.

3.3.7.2 An alemite grease fitting, or equivalent, shall be installed to lubricate the lower radial bearing.

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3.3.7.3 All thrust bearing housings shall be so designed that the bearings will be adequately lubricated immediately when the shaft starts to turn. This shall be accomplished by adequate sealing of the bearing housings so that, regardless of the length of time the units are idle, an adequate supply of lubricant will remain in the bearing housings.

3.3.8 Threaded parts such as bolts, studs and nuts shall conform to Handbook H28.

3.3.8.1 Fine threads may be used for the set ends of studs and in the nut end in cases where special double width cap nuts are proposed any open type hexagonal head nuts shall be of standard size and shall have coarse threads. Nickel-copper locking wire shall be supplied for any threaded parts not readily accessible such as line bearing retainers and end covers.

3.3.9 Welding shall be in accordance with Standard MIL-STD-278.

3.4 Materials. -

3.4.1 All material used in the construction of the pumps and associated equipment shall be of the quality best suited for the purpose intended, and shall be strictly in accordance with the material specifications.

3.4.2 The use of materials differing in properties from those specified for a given application will be considered when the contractor shows the necessity for such substitution, the material proposed is readily obtainable, and provided that the material is satisfactory to the bureau or agency concerned.

3.4.3 Where no specifications are specified herein for the material for certain parts, the material used shall be of a type and quality consistent with that specified for associated parts in the pump. The use or substitution of manufacturer's material shall be subject to approval by the bureau or agency concerned and shall be clearly indicated in a list submitted with the master drawings.

3.4.4 Materials for the pumps shall be in accordance with table I.

Table I - Materials

Name of part	Material	Specification
Casings, bearing, brackets and caps	Gun metal	MIL-M-16576
	Copper-nickel alloy (70-30)	MIL-C-20159
	Aluminum bronze, class 2	MIL-B-16033
	"Worthite" or equivalent	
Shafts	Nickel-copper-aluminum alloy	QQ-N-286
	"Worthite" or equivalent	
Shaft sleeves	Nickel-copper-aluminum alloy (minimum 265 Brinell)	QQ-N-286
	"Worthite" or equivalent	
Impellers	Nickel-copper alloy	QQ-N-288
	Copper-nickel alloy (70-30)	MIL-C-20159
	Gun metal	MIL-M-16576
	"Worthite" or equivalent	
Diaphragms	Gun metal	MIL-M-16576
	Copper-nickel alloy (70-30)	MIL-C-20159
	Hydraulic bronze	MIL-B-16444
	"Worthite" or equivalent	
Lantern rings, glands and throat bushings	Gun metal	MIL-M-16576
	Copper-nickel alloy (70-30)	MIL-C-20159
	"Worthite" or equivalent	
Bearings (sleeve)	Bearing bronze	MIL-B-16261
	"Bearium" metal	

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3.5 Motors. -

3.5.1 Al direct current (d.c.) motors for driving pumps shall be of the types and characteristics specified and shall conform to Specification MIL-M-17413 except as modified herein:

Service	Navy A.
Volts	Varying as specified (see 6.1).
Speed classification	Adjustable varying.
Duty	Continuous and intermittent (see table II).
Enclosure	Dripproof protected.
Mounting	Vertical, shall down
Ambient temperature	50° C.
Winding	Stabilized shunt.
lnsulation	Class B.
Bearings	Ball.

3.5.2 Motors shall be sized so as not to be overloaded at any possible operating point of the pumps up to the maximum speed specified (see 6.1).

3.5.2.1 The motors shall conform to the duty ratings shown in table II.

Table II - Duty rating of motors.

Horsepower [1]	Voltage	Speed [2]	Duty
Maximum	Nominal	Maximum	10 minutes out of 30 minutes
Maximum	70 percent of nominal	Maximum	5 minutes
73 percent of maximum	Nominal	87 percent of maximum	Continuous
73 percent of maximum	80 percent of nominal	87 percent of maximum	30 minutes
73 percent of maximum	70 percent of nominal	87 percent of maximum	10 minutes
36. 5 percent of maximum	142 percent of nominal	Full field	30 minutes

[1]The maximum horsepower (hp.) is the maximum hp. required by the pump under any condition of operation specified (see 6.1).

[2]The maximum speed is the maximum revolutions per minute (r.p.m.) required by the pump to meet all conditions specified (see 6.1).

3.5.3 D.c. motors shall be mechanically capable of operating at a speed of 10 percent above the setting of the overspeed trip, if one is provided. If no overspeed trip is provided, the motors shall be mechanically capable of operating at a speed of 10 percent above the hot no-load speed at the maximum voltage specified with 2 field resistance equal to 110 percent of the rated resistance value of the field rheostat. The overspeed trip, if provided, shall be set at approximately 10 percent above maximum rated speed.

2.5.4 Each d.c. motor shall be equipped with a reliable and accurate electric continuous speed indicator apparatus consisting of a small permanent magnet type alternator permanently mounted on the top of the motor, and a speed Indicating meter suitable for installation remote from the motor. A terminal box shall be provided in the motor for connecting cable from alternator to meter, The cable need not be supplied. The meter shall be suitable for bulkhead mounting and shall have a clear reading 90-degree arc scale not less than 3-3/4 inches long which will give accurate speed readings from 150 degrees to 3600 r. p. m. The speed indicator device and overspeed trip device may be combined in one unit, if desired.

3.5.5 Motors shall be capable of being started over the entire voltage range specified (see 6.1), assuming at the instant of starting, the motor need supply only sufficient torque to start the pump rotor.

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3.5.6 The use of wrought aluminum alloy motor parts will be acceptable. Alloy shall be in accordance with Specification QQ-A-327 and shall be Alcoa No. 615-T, 615-W, 535-T, or 535-W, or equivalent.

3.5. Armature slots shall be skewed with respect to the edge of the pole pieces to reduce magnetic noise.

3.5.8 All traces of brush chatter and "end slop" shall be eliminated.

3.5.9 Motors shall be designed for precision balance in accordance with Specification MIL-M-17413.

3.5 10 Ball bearings conforming to Specifications FF-B-171 and MIL-B-17931 shall be used.

3.6 Controllers. -

3.6.1 D.c. motor controllers shall be of the characteristics specified (see 6.1) and shall conform to Specification MIL-C-2174 except as modified herein:

Performance	Semi-automatic.
Function	Motor starting and speed control.
Duty	Same as motor.
Enclosure	Dripproof.
Master switches	Remote, spray tight, START-STOP push buttons.
Protection	LVP no overload protection
Ambient temperature	50°C.
Acceleration	Time delay.

3.6.2 Field rheostats in accordance with Specification MIL-R-15109 shall be furnished as required to provide motor speed adjustment, over the required speed range.

3.6.3 Each controller shall include a double winding full field relay designed to assure full field operation at voltages above nominal, and at lower voltages when load is heavy. At the nominal voltage approximately 65 percent of the ampere turns at which the contacts of the relay opens shall be in the shunt coil connected across the line, and 35 percent of the ampere turns at which the relay opens shall be in a coil connected in series with the armature circuit.

3.6.4 Motor controllers shall not exceed 15 cubic feet in size and shall weigh not more than 600 pounds.

3.7 Identification plates. -

3.7.1 Identification plates shall be furnished on each pump and shall be type A or B in accordance with Specification MIL-P-15024, except that identification plates of plastic or aluminum are not acceptable.

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3.7.2 Identification plates shall be secured to equipment with corrosion-resistant metallic screws.

3.7.3 Pump identification plates shall contain the following data:

(a) Manufacturer's name.
(c) Manufacturer's model or type and size.
(c) Service application (trim and drain).
(d) Manufacturer's serial number.
(e) Salient design characteristics:
    (1) Capacity range in gallons per minute.
    (2) Total head range in feet.
    (3) Speed range of shaft in r.p.m.
    (4) Maximum brake horsepower.
    (5) Test pressure.
(f) Contract order number (and item number for multiple unit orders).
(g) Bureau of Ships stock number.
(h) Section for inspector's stamp.

3.7.4 Each driving unit and each accessory unit shall have an identification plate in accordance with the applicable equipment specification.

3.8 Painting. - All external unmachined surfaces of ungalvanized ferrous metal parts shall be thoroughly cleaned and coated with one coat of pretreatment, Specification MIL-C-15328, and one coat of zinc chromate primer, Specification JAN-P-735, or one coat of red lead primer, Specification MIL-P-17545, followed by a finish coat of light gray equipment enamel, Specification MIL-E-15090. Painting of external surfaces of nonferrous parts of pumps will not be required.

3.9 Drawings -

3.9.1 Drawings shall be in accordance with type I of Specification MIL-D-963.

3.9.2 Outline drawings shall be furnished covering complete pump units including pump, driver, and attached accessories. Separate outline drawings shall be furnished for each contract or order and for each different pump unit on a contract or order.

3.9.2.1 Outline drawings shall show the radius of gyration and center of gravity of the pump and motor unit.

3.9.2.2 Performance curves on preliminary outline drawings as originally submitted for approval may be calculated curves and shall be so indicated. The complete performance curves on the validated outline drawings shall be based on the performance acceptance tests and shall be so indicated.

3.9.2.3 Pump master drawings shall include sectional assembly, complete list of material and details of all parts. The sectional assembly and list of material with references to detail drawings should be on a single sheet; and details shall be on one or more additional sheets. Several details should be shown on each pump master drawing. Details of single pump parts on size A sheets, resulting in a multiplicity of drawing sheets, should be avoided.

3.9.2.4 Subassemblies furnished as repair parts in lieu of or in addition to individual repair parts shall be recorded on the related outline drawings by reference notes or as additional items on the "Lint of Material" of the sectional assembly drawing.

3.9.2.5 Drawings for drivers and associated equipment shall be in accordance with the specifications for those equipments.

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3.9.2.6 The sectional assembly drawing shall show all assembly clearances.

3.9.2.7 Drawings submitted with bids are not required to be m master drawing format.

3.10 Technical manuals. - Technical manuals shall be in accordance with type III of Specification MIL-M-15071, unless otherwise specified in the contract or order. A separate manual shall be furnished for each different pump unit except in special cases when manuals covering more than one pump may be approved. Preliminary technical manuals submitted for approval shall include all proposed sections complete.

3.10.1 Preliminary technical manuals, containing the actual pump performance curves, shall be packed with each pump in the event final manuals are not prepared.

3.11 Repair parts and tools. -

3.11.1 Onboard repair parts of the types and quantities listed in table III shall be furnished for each submarine as required (see 6.1).

Table III - Onboard repair parts.

Name of part	Quantity
Thrust bearing with locknut and washer	1
Radial bearing assembly	1
Lockwire	Sufficient for one pump
Stuffing box packing for pump	1 set
Coupling wearing parts	1 set
Coupling locknut and washers	1
Shaft sleeve	1
Ball bearing seals	1 set
Throat bushing	1
Lantern ring	1
Water flinger and key or set screw	1
Fastening for glang halves	1 set

3.11.1.1 Onboard repair parts for the electrical equipment shall be furnished in accordance with the applicable equipment specification.

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3.11.2 Onboard repair parts for stock aboard submarine tenders shall be of the type and quantities listed in table IV.

Table IV - Tender repair parts.

Name of part	Quantity
Pump complete, less motor and controls	1
Rotor assembly, complete	1
Ball bearing with lock washer and nut	1
Sleeve bearing with retainer and retainer flange	1
Lockwire	1
Packing	1 set
Grid member (coupling)	1
Seal member (coupling)	2
Shaft sleeves	2
Guide bushing	1
Lantern ring	1
Water flinger	1
Diaphragm, undersize	1 set
Thrust washer	1
Half packing gland	2
Locking lever spring	1

3.11.3 Tools .-

3.11.3.1 A complete set of special tools required for the maintenance and repair of the pump units shall be included in each set of onboard repair parts and each set of tender parts.

3.11.3.2 Tools which are available in the Navy Stock List of General Stores, such as common wrenches and standard pullers, will not be required. Standard complete or limited sets of pullers or parts of puller sets in accordance with Specification GGG-P-781 shall be identified as to type and use in notes on the outline or section assembly drawings. Specials tools shall be detailed, and included in the lsit of material. Special tools are defined as those tools not listed in the Navy Stock List of General Stores. (Copies of this stock list may be consulted in the Office of the Government inspector.)

3.11.4 Each box containing onboard repair parts shall contain a list entitled “List of onboard repair parts and tools”. The list shall be in format suitable for use on outline drawings and in the technical manuals. It shall not be less than nominal 8-1/2 by 11 inches in size.

3.11.4.1 The format of the list shall include a heading and columns of data for the items listed. The heading shall include titles and applicable entries as follows:

Reproduced from drawing number _____
Number of ships _____
Application _____
Contract or order number _____
Manufacturer _____
Quantities are for _____ units per ship
The columns shall include:
Piece number _____
Name of part _____
Quantity _____
Drawing number _____
Standard Navy stock number _____
Additional columns may be used as applicable

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3.11.4.2 The list shall be so treated as to be resistant to oil, water and fading.

3.11.5 Stock. - The selection stock numbering and quantities of repair parts for stock of Naval supply activities shall be determined and processed in accordance with Specification MIL-R-15137.

3.11.5.1 The cognizant supply demand control point for repair parts for stock is Submarine Supply Office, Phila. Pa., unless otherwise specified in the contract or order.

3.12 Workmanship. - The equipment, including all parts and accessories, shall be manufactured and finished with first class workmanship in all respects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Qualification tests. -

4.1.1 Qualification tests at a laboratory. - Qualification tests shall be conducted at a laboratory designated by the Bureau of Ships. The head, capacity and other characteristics of units for qualification tests shall be as specified by the Bureau of Ships.

4.1.2 In addition to the tests at the laboratory, pumps may be required to pass the following shipboard service test:

(a) Two or more pumps of the general design and general operating characteristics shall have given satisfactory service onboard a Naval ship for a period of at least two years and must still be in service on such ship at the time qualification is granted.

4.1.3 The qualification tests shall adequately demonstrate the ability of the pump to handle the specified capacities at the maximum specified temperature of the liquid and the minimum suction head or maximum suction lift or vacuum. Sufficient data shall be taken during the tests to prepare pump characteristic curves as required herein. Test data and curves shall be complete over the entire range of capacities from shut-off to as near free delivery as possible and shall be obtained for at least two speeds on both series and parallel operations. The following test curves shall be prepared:

(a) Capacity versus total head,
(b) Capacity versus pump efficiency.
(c) Capacity versus brake horsepower.

4.1.4 Endurance test. - The pump shall be operated with impeller at maximum speed for 24 hours continuous at a capacity as near free delivery as possible and for 12 hours with impeller, in series at maximum speed at a capacity as near shut-off as possible. Failure of any part of the pump during the endurance test shall be considered as failure of the pump to quality. The pump shall complete the endurance test without excessive noise, vibration or wear of parts.

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4.2 Shop tests,-

4.2.1 Pumps shall withstand a hydrostatic test pressure of 1,000 pounds per square inch gage. Small "weeps" of less than five drops per minute at gasketed joints during the hydrostatic test shall not be cause for rejection of the pumps, No leakage is permissible at shut-off pressure.

4.2.2 Each pump, shall be tested in the shop of the manufacturer or contractor at designed maximum operating speed by a continuous nonstop run of at least 30 minutes. This test shall be conducted in the presence of the Government inspector, who shall check operation and smoothness of running. While running at this speed the pump need not be loaded except as necessary to prevent injury.

4.2.3 Each completely assembled pumping unit shall be tested for airborne noise, structure-borne vibration, and pump discharge pressure pulsations in accordance with the addendum to Purchase Specifications for Machinery - Noise and Vibration. This test shall be conducted in the presence of the Government inspector at the plant of the manufacturer or the contractor. These tests may be made during the 30-minute test run specified in 4.2.2. Results of these tests shall be reported promptly to the Bureau of Ships, Code 549.

4.3 Performance acceptance tests. -

4.3.1 It shall be the responsibility of the prime contractor to insure that tests required on assembled units are made.

4.3.2 Motors and controllers shall be tested as required by the applicable equipment specification,

4.3.3 One pump of each type and size on the contract or order, as selected by the Government inspector, complete with the motor and controller, shall be given a test to determine the overall power consumption of the unit under the specified or guaranteed conditions. This test may be conducted in the shop of the main contractor or subcontractor as arranged with the bureau or agency concerned. The Bureau of Ships may consider waiver of overall power tests on pumps if identical units with identical guarantees have been previously tested.

4.3.4 Performance acceptance tests shall ordinarily be conducted at the same time as, and shall include, the reclined operation tests as specified for the driving unit. For inclined operation tests the unit shall be run at maximum operating speed, but need not be fully loaded.

4.3.5 Performance acceptance tests shall adequately demonstrate the ability of the pump to handle its rated capacity of specified liquid at the minimum suction head or maximum suction lift or vacuum, as applicable.

4.3.6 Sufficient data shall be taken during the tests to prepare pump characteristic curves as specified in 4.3.7. In all cases the test data shall be corrected to the specified operating conditions as to voltage, temperatures, specific gravity, suction head or lift, and vacuum, as set forth in the contract or order, such conditions shall be clearly shown on the data sheets. Test data and curves shall be complete over the entire range of capacities from shut-off to as near free delivery as possible.

4.3.7 The following test curves shall be supplied for constant speeds as required to obtain the ratings specified (see 6.1):

(a) Capacity versus total head.
(b) Capacity versus pump efficiency.
(c) Capacity versus brake horsepower.
(d) Capacity versus electrical horsepower input.

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4.3.8 The bureau or agency concerned may, when requested, authorize separate tests of the driving unit and driven unit, at the plants of the different manufacturers, and will accept overall power results and characteristic curves computed from the results therefrom. This should not be construed in any way to release the primary contractor from an operating run on one completely assembled unit from each contract. This operating test shall be run with the unit fully loaded and shall demonstrate all requirements of the specifications as to thrust, governing, balance, and lubrication, at maximum speed.

4.3.9 Complete test reports, including test data on both pumps and driving units, shall be assembled and distributed as follows:

(a) Three to the Bureau of Ships (to be forwarded via the coginzant inspector).
(b) One for primary district inspection office.
(c) One for branch inspection office (if any).
(d) Two to each supervisor of shipbuilding (one for supervisor's files, the other for the shipbuilder).
(e) One to the Commander of each shipbuilding yard concerned, for Navy yard built ships, when specified in the contract or order.

4.3.10 One complete pumping unit of each type, design and size shall be subjected to the high impact shock test specified in Specification MIL-S-901 as required by 3.2.8 herein.

4.3.11 Shock tests will be conducted at the manufacturer's plant. Where a manufacturer is unable to conduct these tests at his own plant, he may arrange to have them conducted at a commercial laboratory, or Government laboratory suitably equipped to conduct tests. Where shock tests are conducted at a Government laboratory, copies of the applicable master drawings shall accompany the units.

4.3.12 Equipment previously accepted will not be resubmitted for shock test except when evidence of low shock resistance develops in the units installed. The bureau or agency concerned will initiate action with the manufacturer for correction of deficiencies, and reserves the right to require shock tests at any time.

4.3.13 All pump units less tachometer shall be shock tested with drivers unless otherwise approved by the bureau or agency concerned. Pumps shock tested with one driver will not be required to be shock tested again when supplied with a different driver. Prime movers are subjected to shock tests in accordance with the applicable equipment specification.

4.3.14 One pump of each manufacturer's type and each nominal size shall be subjected to shock tests, Compliance with shock resistance requirements may be covered by test of a similar unit subject to approval by the bureau or agency concerned.

4.3.15 The pump unit less tachometer shall be considered to have failed to pass the shock tests in event of any of the following:

(a) Breakage of any parts, including mounting bolts.
(b) Appreciable distortion or dislocation of any part, such as shaft, mounting feet, bearings.
(c) A mechanical unbalance of more than two times the amplitude of unbalance measured prior to tests at rated speed, or more than that permitted by the specifications for the driver.

4.3.16 Pump units which have been subjected to the high-impact shock test and have failed to conform to the requirements will not be acceptable.

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4.3.17 Pump units which have been subjected to the high-impact shock test and have successfully passed this test shall be considered acceptable for such service as the bureau or agency concerned may authorize. Such equipment will be returned by the Government to the contractor for reconditioning as follows:

(a) Minor deformations of mounting flanges shall be corrected.
(b) Minor deformations affecting alignment shall be corrected.
(c) All bearings shall be replaced.
(d) Each part shall be carefully examined by the contractor and any part which he considers substandard shall be replaced.

Minor deformations shall be defined as those which do not cause unqualified rejection of the design under high-impact shock test but which are in excess of the design dimensional tolerances specified on the applicable pump drawings.

4.4 Inspection of pump repair parts. - The pump repair parts required by 3.11.1 shall be inspected to assure interchangeability with the similar parts in the assembled pumps. Ordinarily material analyses and physical test specimens will not be required, but the Government inspector shall employ such procedures as will assure the delivery of parts made of the same materials by the same processes as for the assembled pumps.

4.5 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with Genera] Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. - Pumps and repair parts shall be prepared for delivery for domestic shipment - immediate use; domestic shipment and storage; or for overseas shipment in accordance with Specification MIL-P-16789 (see 6.1).

6. NOTES

6.1 Ordering data. - Procurements documents should specify the following:

(a) Title, number, and date of this specification.
(b) Performance ratings.
(c) Liquid handled.
(d) Specific gravity of liquid handled.
(e) Electrical characteristics as required by applicable motor and controller specifications.
(f) If repair parts boxes are required, specify type (metal or wood) (see 5.1).
(g) Whether for domestic shipment - immediate use; domestic shipment and storage; or for overseas shipment (see 5.1).
(h) That the following data is required With bids:
    (1) Sectional assembly drawing of pump.
    (2) A drawing of the unit showing overall dimensions.
    (3) Pump characteristic curves.
    (4) Horsepower of driver (continuous duty rating at nominal voltage and at maximum specified voltage).
    (5) Weight of complete unit.
    (6) The manufacturer and descriptive data of the driver and associated accessories, such as controller and tachometer.

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6.2 Qualification. - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL-21343, whether or not such products have actually been so listed by that date.

6.2.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D.C.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Bureau of Ships
(Project 4320-0002Sh)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEM PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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